

PMEG2010AEB

20 V, 1 A low VF Schottky barrier rectifier

6 August 2026 Product data sheet

1. General description

Planar Schottky barrier rectifier with an integrated guard ring for stress protection, encapsulated in a SOD523 (SC-79) ultra small and flat lead Surface-Mounted Device (SMD) plastic package.

2. Features and benefits

- Forward current: 1.0 A
- Reverse voltage: 20 V
- Ultra low forward voltage
- Ultra small SMD package

3. Applications

- Low voltage rectification
- High efficiency DC/DC conversion
- Voltage clamping
- Inverse-polarity protection
- Low power consumption applications

4. Quick reference data

Table 1. Quick reference data

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_R	reverse voltage			-	-	20	V
V_F	forward voltage	I_F = 1 A; T_{amb} = 25 °C		-	510	620	mV
I_F	forward current	$T_{sp} \leq$ 55 °C		-	-	1	A

5. Pinning information

Table 2. Pinning information

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	K	cathode	 SC-79 (SOD523)	 sym001
2	A	anode		

6. Ordering information

Table 3. Ordering information

Type number	Package		
	Name	Description	Version
PMEG2010AEB	SC-79	plastic, surface-mounted package; 2 leads; 1.2 mm x 0.8 mm x 0.6 mm body	SOD523

7. Marking

Table 4. Marking codes

Type number	Marking code
PMEG2010AEB	L6

8. Limiting values

Table 5. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134).

Symbol	Parameter	Conditions		Min	Max	Unit
V_R	reverse voltage			-	20	V
I_F	forward current	$T_{sp} \leq 55$ °C		-	1	A
I_{FRM}	repetitive peak forward current	$t_p \leq 1$ ms; $\delta \leq 0.5$		-	3.5	A
I_{FSM}	non-repetitive peak forward current	square-wave pulse; $t_p = 8$ ms		-	6	A
T_j	junction temperature		[1]	-	150	°C
T_{amb}	ambient temperature		[1]	-65	150	°C
T_{stg}	storage temperature			-65	150	°C

[1] For Schottky barrier diodes thermal runaway has to be considered, as in some applications the reverse power losses P_R are a significant part of the total power losses. Nomograms for determining the reverse power losses P_R and $I_{F(AV)}$ rating will be available on request.

9. Thermal characteristics

Table 6. Thermal characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient	in free air	[1] [2]	-	-	400	K/W
$R_{th(j-sp)}$	thermal resistance from junction to solder point		[2] [3]	-	-	75	K/W

[1] Refer to SOD523 (SC-79) standard mounting conditions.
[2] For Schottky barrier diodes thermal runaway has to be considered, as in some applications the reverse power losses P_R are a significant part of the total power losses. Nomograms for determining the reverse power losses P_R and $I_{F(AV)}$ rating will be available on request.
[3] Solder point of cathode tab.

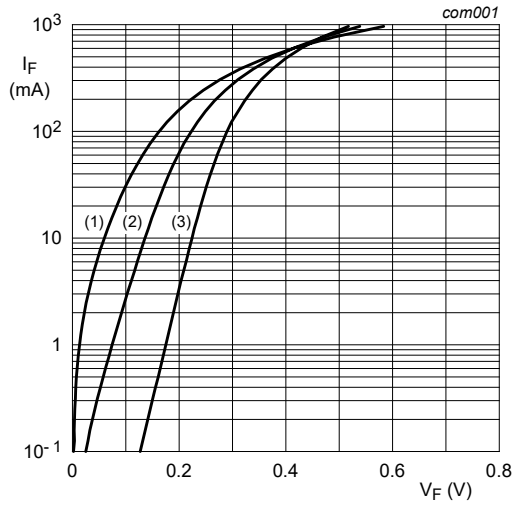
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10. Characteristics

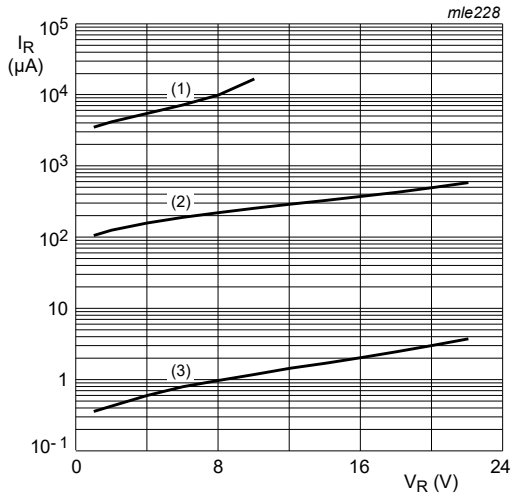
Table 7. Characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_F	forward voltage	I_F = 0.1 mA; T_{amb} = 25 °C		-	30	60	mV
		I_F = 1 mA; T_{amb} = 25 °C		-	80	110	mV
		I_F = 10 mA; T_{amb} = 25 °C		-	140	190	mV
		I_F = 100 mA; T_{amb} = 25 °C		-	230	290	mV
		I_F = 1 A; T_{amb} = 25 °C		-	510	620	mV
I_R	reverse current	V_R = 10 V; pulsed; $t_p \leq$ 300 µs; $\delta \leq$ 0.02; T_{amb} = 25 °C		-	0.17	0.6	mA
		V_R = 20 V; pulsed; $t_p \leq$ 300 µs; $\delta \leq$ 0.02; T_{amb} = 25 °C		-	0.32	1.5	mA
C_d	diode capacitance	V_R = 1 V; f = 1 MHz		-	19	25	pF



(1) T_{amb} = 85 °C
(2) T_{amb} = 25 °C
(3) T_{amb} = -40 °C

Fig. 1. Forward current as a function of forward voltage; typical values

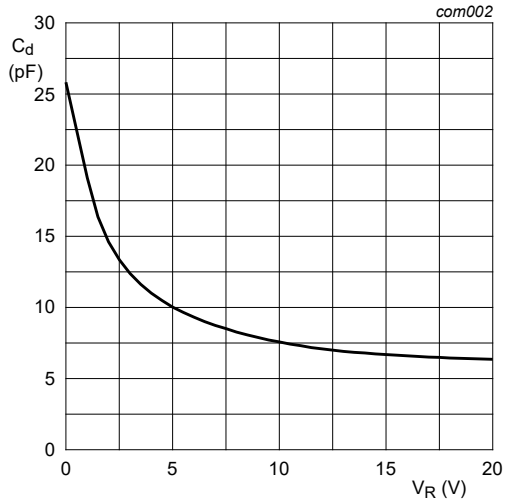


(1) T_{amb} = 85 °C
(2) T_{amb} = 25 °C
(3) T_{amb} = -40 °C

Fig. 2. Reverse current as a function of reverse voltage; typical values

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f = 1 MHz; T_{amb} = 25 °C.

Fig. 3. Diode capacitance as a function of reverse voltage; typical values

11. Package outline

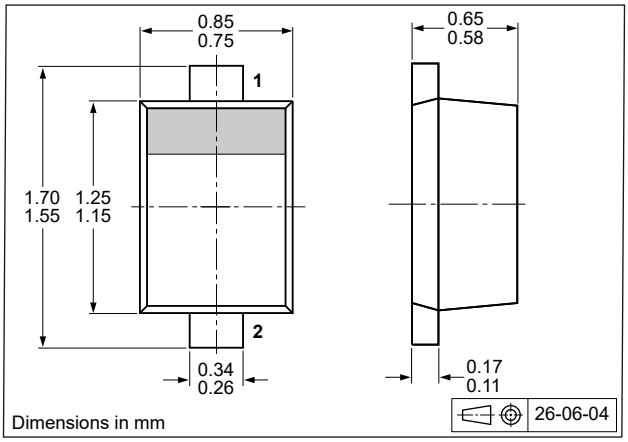


Fig. 4. Package outline SC-79 (SOD523)

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12. Soldering

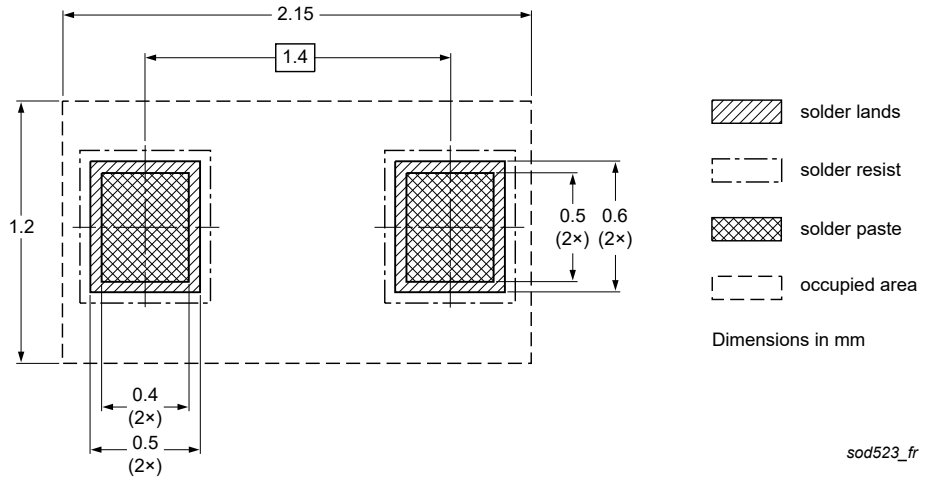


Fig. 5. Reflow soldering footprint for SC-79 (SOD523)

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13. Revision history

Table 8. Revision history

Data sheet ID	Release date	Data sheet status	Change notice	Supersedes
PMEG2010AEB v.3	20260806	Product data sheet	-	PMEG2010AEB v.2
Modifications:	• Aligned SOD523 package outline drawing			
PMEG2010AEB v.2	20221001	Product data sheet	-	PMEG2010AEB v.1
PMEG2010AEB v.1	20190924	Product data sheet	-	-

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Data sheet status

Document status [1][2]	Product status [3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

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[2] The term 'short data sheet' is explained in section "Definitions".
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Date of release: 6 August 2026